POLITECNICO MILANO 1863

BASI DI DATI 1 – Anno Accademico 2024-2025

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Appello 11/06/2025, **DURATA: 2 ore**

Part 1

Exercise 1 – SQL (13pt)

Consider the following database:

TRAIN(Number, FirstClassCoaches, SecondClassCoaches)
STOP(TrainNumber, StationName, Track, ArrivalTime, DepartureTime)
STATION(Name, TracksNumber, DailyPassengers)

A. Specify in SQL the creation of the STOP table, defining all necessary constraints. (1 pt)

B. Find the longest train passing through Lambrate station. (4 pts)

C. Find the express trains that connect only stations visited by more than 100,000 daily passengers. (4 pts)

D. Find all direct trains connecting Milano Centrale (departure) to Roma Termini (arrival) with at most two intermediate stops and within 5 hours. (4 pts)

Exercise 2 – Formal languages (6pt)

Write query C in optimized Relational Algebra (3pt) and Tuple Relational Calculus (3pt).

PART 2

Exercise 3 – Theory question (1pt):

Briefly describe how conceptual, logical and physical design differ.

Exercise 4 – Conceptual and logical design (13pt)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity. Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*
- *Arrange the conceptual project and the logical project so that they can be esaily compared (on two consecutive pages).*

A court manages data on ongoing criminal proceedings. Each trial has one or more prosecutors and one or more defendants, each associated with a defence counsel; the trial may be either first instance or second instance (appeal), in which case the first instance trial that preceded it is known.

Each trial has a panel of judges, headed by a president. The members of the panel and the prosecutors are judges, employees of the court; they are characterised by a name and a code. Defence counsels are registered with a professional roll, and are therefore characterised by a roll number. Each defendant is tried for various offences; he has a name and an identity document. Some defendants are free, others are detained in prison.

The trial is held in several hearings; each hearing has a date and is held in a court of law. At each hearing, witnesses are summoned, characterised by a name and an identity document; witnesses may also be detained. At each hearing, the system prints a list of detainees (defendants and witnesses) to be brought to court, which is communicated to the police headquarters.

At the end of the trial, the sentence is delivered: each defendant may be acquitted or sentenced to prison.